

News, Sentiment and Confidence in Fluctuations

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DEEQA mini course
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Plan of the three sessions

- ▶ Session 1 : A review of the (my) research on "news shocks"
- ▶ Session 2 : Review again + focus on some other work if time permits (Jaimovitch & Rebelo, Uribe & Schmit-Grohé, Barsky & Sims)
- ▶ Session 3 : A "gains from trade" approach to news, sentiment and confidence.

1. Motivations

- ▶ Long lasting question in macroeconomics: what are the sources of economic fluctuations?
- ▶ Macroeconomic fluctuations may be not driven only by current developments in the economy but may also be influenced by perceptions of future developments,
- ▶ Perception = news, sentiment, confidence.

1. Motivations

- ▶ The 1999-2001 boom bust cycle in the US is one example which may fit this idea.
- ▶ The 1997 Asian Crisis is another example.
- ▶ The subprime market collapse of 2007 is yet another example.
- ▶ Changes in expectations as opposed to current changes in opportunities or preferences.

1. Motivations

- ▶ In this talk, I want to review the work my coauthors and myself have done to substantiate this idea that changes in perceptions, and in particular news shocks, matter.
- ▶ By *news shocks*, I mean anything changes expectations *without affecting current preferences, technologies nor policies*.
- ▶ I will be a mix of theory, data and quantitative macro.

Outline of the Talk

1. Motivations
2. Facts
3. Why are those facts challenging for theory?
4. Theory in simple models
5. Application 1: A theory of recessions
6. Application 2: A theory of international comovements
7. Adding heterogeneity
8. Application 3: Non inflationary business cycles
9. Fiscal Policy

Outline of the Talk

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2.1. How to observe news shocks

- ▶ Expectations (and expectation revisions) are not in the informational set on the econometrician
- ▶ Two possibilities:
 1. Narrative approach: looking at episodes
 2. More systematic approach: exploit some variability in the data.

2.2. Narrative Approach

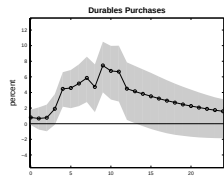
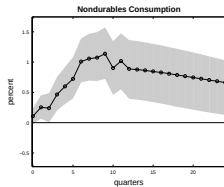
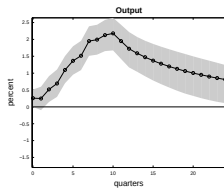
- ▶ Take an episode that is known as changing agents expectations, and then look at the consequences (First Gulf war (Blanchard aer-1998, Maastricht treaty ratification (Portier wp-2005)
- ▶ Nice recent work of Mertens & Ravn wp-2008 on announced *versus* surprising U.S. tax shocks

2.2. Narrative Approach

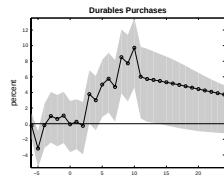
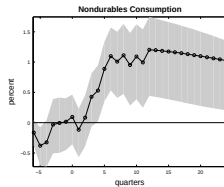
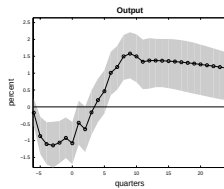
Mertens & Ravn aer-2012, ej-2011, red-2011

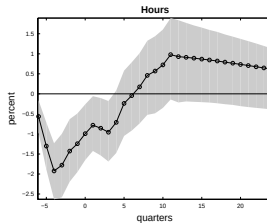
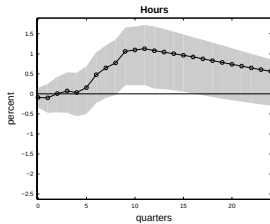
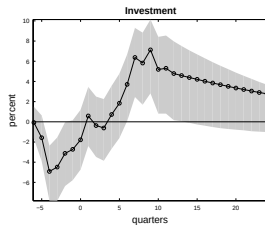
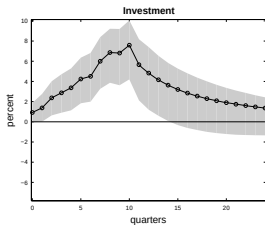
- ▶ Romer & Romer aer-2010 identify 49 significant legislated federal tax acts in the period 1947-2006 and a total of 104 separate changes in tax liabilities.
- ▶ For each of this change, one can observe the implementation lag (*“difference in timing between dates at which the tax legislations were signed by the President and the dates at which the tax liability changes were to be implemented”*)
- ▶ 50% of episodes are anticipated and the median implementation lag of these legislations is 6 quarters.
- ▶ Results: - Agents do react to anticipated tax cut - They do so in in different manner compared to unexpected ones

Unanticipated Tax Shock



Anticipated Tax Shock





2.3. Systematic Approach (Beaudry & Portier 2007)

- ▶ The idea is to look at some variables that capture agents expectations
- ▶ Typically asset prices.
- ▶ Let's restrict to possible information about future productivity.
- ▶ Do we find in asset prices data news about future total factor productivity (TFP)? How does the economy react to those news?

2.3. Systematic Approach (Beaudry & Portier aer-2007)

- ▶ Simple bi-variate system.
- ▶ Measured total factor productivity TFP_t , and a forward looking economic decision variable X_t .
- ▶ The only characteristic of X_t that is important for our argument is that it be a jump variable, that is, a variable that can immediately react to changes in information without lag (stock price, interest rate, even consumption).

2.3. Systematic Approach (Beaudry & Portier 2007)

- ▶ We look at the response of the TFP and the stock market X to a long run shock to TFP (“the” technology shock, called $\tilde{\epsilon}_1$)
- ▶ We look at the response of the TFP and the stock market X to a stock market innovation (“the” non-technology shock, called ϵ_2)
- ▶ Standard theory suggest $\epsilon_2 \perp \tilde{\epsilon}_1$.
- ▶ We show that data suggest $\epsilon_2 = \tilde{\epsilon}_1$.

2.3. Systematic Approach (Beaudry & Portier aer-2007)

The VARs

- ▶ We consider two alternative representations of the Bivariate VAR with orthogonalized errors:

$$\begin{pmatrix} \Delta TFP_t \\ \Delta X_t \end{pmatrix} = \Gamma(L) \begin{pmatrix} \epsilon_{1,t} \\ \epsilon_{2,t} \end{pmatrix}, \quad (1)$$

$$\begin{pmatrix} \Delta TFP_t \\ \Delta X_t \end{pmatrix} = \tilde{\Gamma}(L) \begin{pmatrix} \tilde{\epsilon}_{1,t} \\ \tilde{\epsilon}_{2,t} \end{pmatrix}, \quad (2)$$

- ▶ short run restriction: $\Gamma_{0(1,2)} = 0$ (ϵ_2 has no contemporaneous impact on TFP)
- ▶ long run restriction: $\tilde{\Gamma}(1)_{(1,2)} \left[= \sum_{i=0}^{\infty} \tilde{\Gamma}_i \right] = 0$ ($\tilde{\epsilon}_2$ has no long run impact on TFP)

2.3. Systematic Approach (Beaudry & Portier aer-2007)

Prediction of a simple RBC model with technology and preference shocks

$$U = E_0 \sum_{t=0}^{\infty} \beta^t \left[\log C_t - \Lambda_t \frac{L_t^\sigma}{\sigma} \right]$$

$$\Lambda_t = e^{\eta_{2,t}}$$

$$K_{t+1} = I_t$$

$$Y_t = \theta_t K_t^\gamma L_t^{1-\gamma}$$

$$\theta_t = \theta_{t-1} e^{\eta_{1,t}}$$

2.3. Systematic Approach (Beaudry & Portier aer-2007)

Prediction of a simple RBC model with technology and preference shocks

- The model solution can be written as

$$\begin{pmatrix} \Delta TFP_t \\ \Delta p_t^b \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ (1-\gamma)\frac{1}{1-\gamma L} - 1 & -\frac{(1-L)(1-\gamma)^2}{\sigma(1-\gamma L)} \end{pmatrix} \begin{pmatrix} \eta_{1,t} \\ \eta_{2,t} \end{pmatrix} \quad (1)$$

- This model implies $\epsilon_1 = \eta_1$, $\epsilon_2 = \eta_2$, $\tilde{\epsilon}_1 = \eta_1$ and $\tilde{\epsilon}_2 = \eta_2$
- In particular, this type of model implies that $\epsilon_2 \perp \tilde{\epsilon}_1$.

2.3. Systematic Approach (Beaudry & Portier aer-2007)

Prediction of a simple RBC model with news shocks

- ▶ Small deviation from the RBC model
- ▶ TFP has both a permanent component – $\bar{\theta}_t$ – and a temporary component – ν_t , and we disregard preference shock.
- ▶ The important additional assumption is that permanent innovation to technology are known to agents 1 period before they actually impact TFP .

$$TFP_t = \bar{\theta}_t + \nu_t$$

$$\bar{\theta}_{t+1} = \bar{\theta}_t + \eta_{1,t}$$

$$\nu_t = \rho \nu_{t-1} + \eta_{2,t}, \quad 0 < \rho < 1$$

2.3. Systematic Approach (Beaudry & Portier aer-2007)

Prediction of a simple RBC model with news shocks

- ▶ The model solution can be written as

$$\begin{pmatrix} \Delta TFP_t \\ \Delta p_t^b \end{pmatrix} = \begin{pmatrix} L & \frac{\gamma(1-L)}{(1-\rho L)} \\ \frac{(1-\gamma)L}{1-\gamma L} - 1 & \frac{(-\rho)(1-L)}{(1-\rho L)} \end{pmatrix} \begin{pmatrix} \eta_{1,t} \\ \eta_{2,t} \end{pmatrix} \quad (2)$$

- ▶ This model implies $\epsilon_1 = \eta_2$, $\epsilon_2 = \eta_1$, $\tilde{\epsilon}_1 = \eta_1$ and $\tilde{\epsilon}_2 = \eta_2$
- ▶ In particular, we have that ϵ_2 is co-linear to $\tilde{\epsilon}_1$.

2.3. Systematic Approach (Beaudry & Portier 2007)

Discussion

- ▶ Important aspect of the two first models: business cycle fluctuations can be decomposed into structurally meaningful supply driven and demand driven components.
- ▶ Demand is orthogonal to TFP and has no long run effect.

2.3. Systematic Approach (Beaudry & Portier aer-2007)

Discussion

- ▶ The news model is different.
- ▶ Before supply shocks actually materialize, the economy already incorporates it.
- ▶ Such a model does not validate a decomposition in terms of demand versus supply effect.
- ▶ In the short run, an anticipated technological improvement (a news shock) looks like a demand effect, while in the long run it looks like a supply effect.

2.3. Systematic Approach (Beaudry & Portier aer-2007)

Data

- ▶ 1948-2000 US data
- ▶ 1960-2000 Japanese data

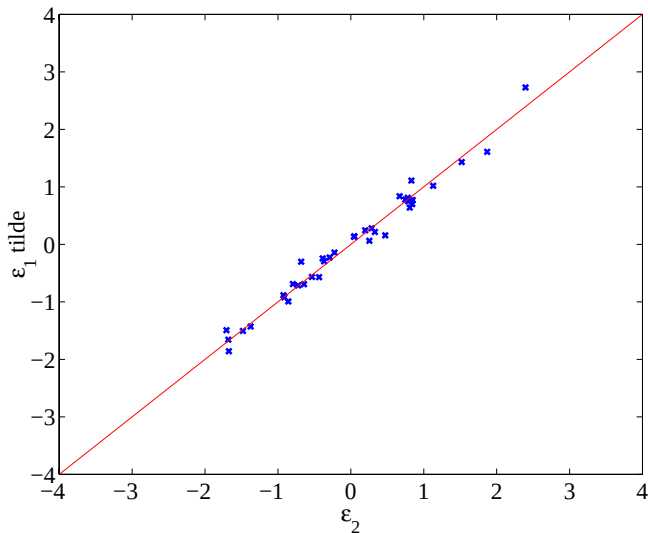
2.3. Systematic Approach (Beaudry & Portier 2007)

Bi-variate VAR

- ▶ We model the joint behavior of TFP and the stock price index.
- ▶ We obtain IRFs and epsilons

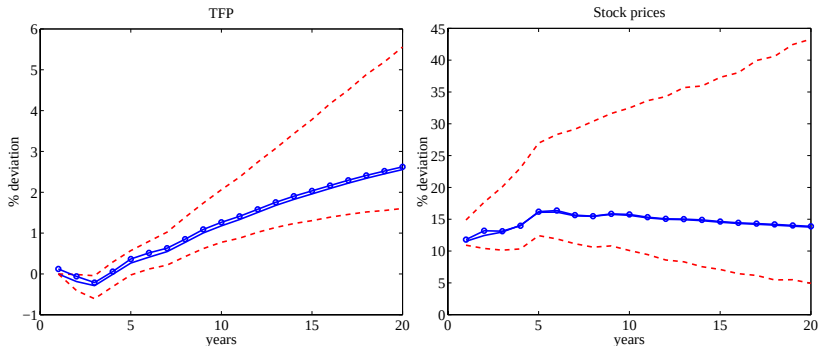
2.3. Systematic Approach (Beaudry & Portier aer-2007)

ϵ_2 Against $\tilde{\epsilon}_1$ – U.S.



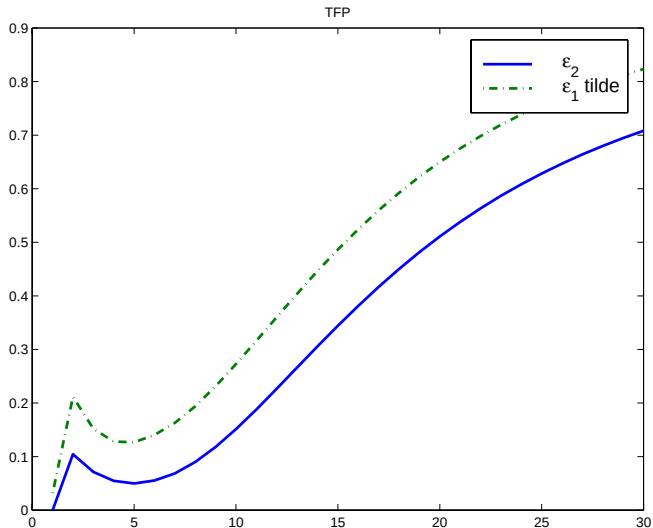
2.3. Systematic Approach (Beaudry & Portier aer-2007)

Impulse Responses to Shocks ϵ_2 and $\tilde{\epsilon}_1$ – U.S



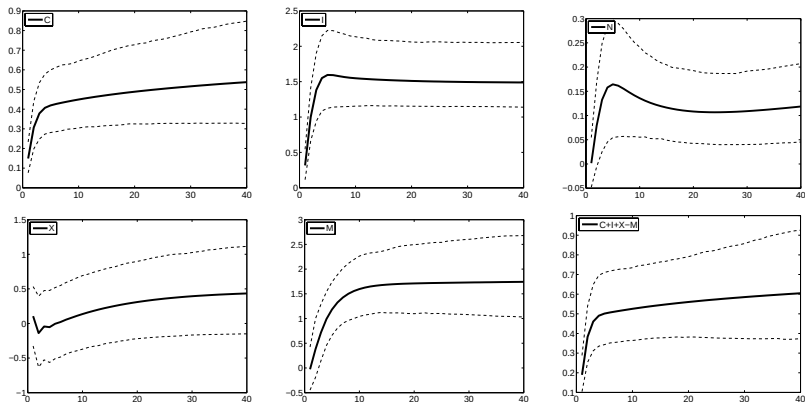
2.3. Systematic Approach (Beaudry & Portier aer-2007)

U.S.A.: Forecast Error Variance of TFP explained by ϵ_2 and $\tilde{\epsilon}_1$



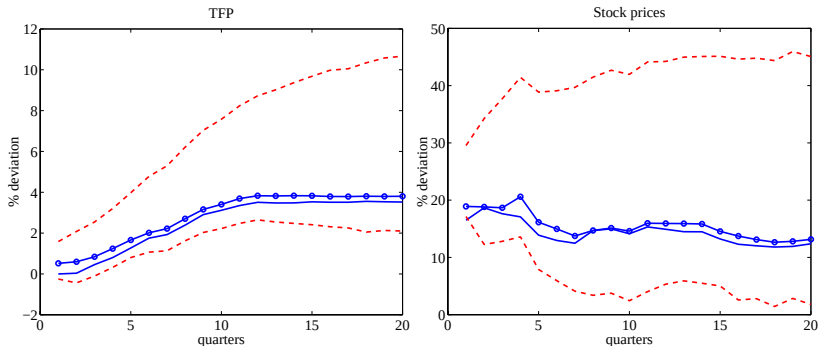
2.3. Systematic Approach (Beaudry & Portier aer-2007)

A news shock triggers an expansion in the US.



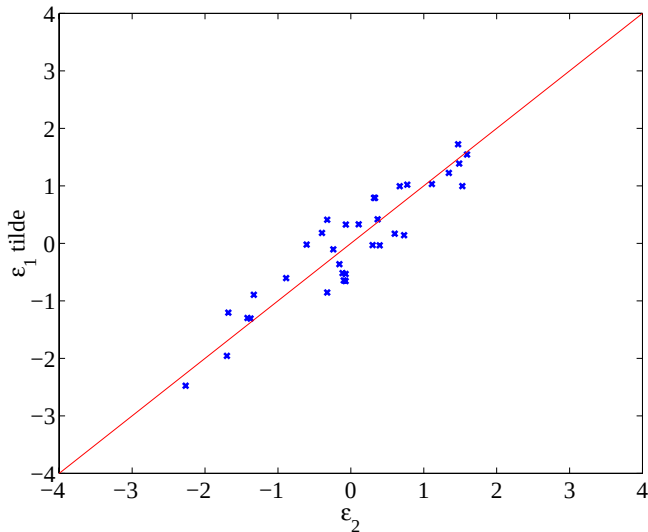
2.3. Systematic Approach (Beaudry & Portier aer-2007)

Impulse Responses to Shocks ϵ_2 and $\tilde{\epsilon}_1$ – Japan



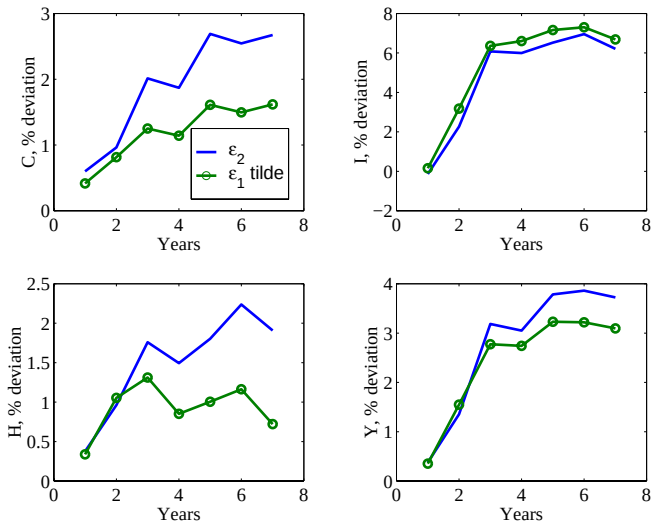
2.3. Systematic Approach (Beaudry & Portier aer-2007)

ϵ_2 Against $\tilde{\epsilon}_1$ – Japan



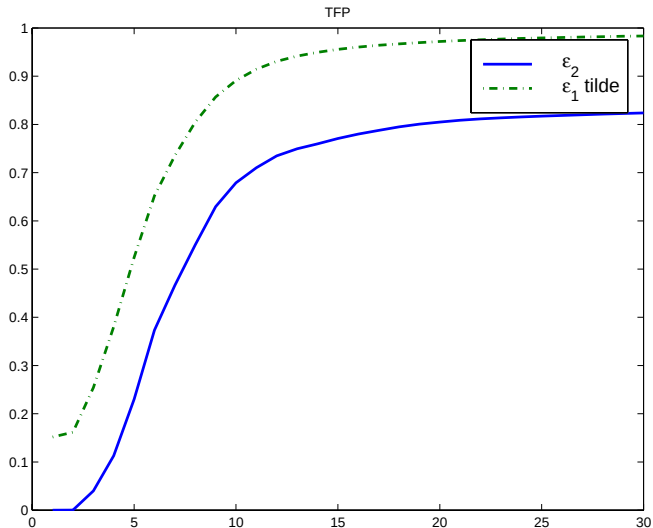
2.3. Systematic Approach (Beaudry & Portier aer-2007)

Response to ϵ_2 and $\tilde{\epsilon}_1$ – Japan



2.3. Systematic Approach (Beaudry & Portier aer-2007)

Japan: Forecast Error Variance of TFP explained by ϵ_2 and $\tilde{\epsilon}_1$



2.4. Summary

- ▶ Some plausible evidence that News shocks are around and that they create business cycle fluctuations
- ▶ So what? Is that a big deal for the standards macro models we know?

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3. Why are those facts challenging for theory?

3.1. At first sight, those episodes cannot be captured by simple neoclassical models

- ▶ Think of an optimal growth model with fixed labor supply.
- ▶ Without shocks to current fundamentals, the economy has to move along the resource constraint

$$C + I = F(K, \bar{L})$$

- ▶ $\rightsquigarrow$ no BC movements are possible: if $C \nearrow$, then $I \searrow$

3.1. At first sight, those episodes cannot be captured by simple neoclassical models

- ▶ If labor supply is not fixed, the impossibility result is not changed if leisure is a normal good:

$$C + I = F(K, L)$$

- ▶ if $C \nearrow$, then $L \searrow$ (because leisure goes up as consumption does), and necessarily $I \searrow$
- ▶ Is this impossibility result general?
- ▶ Let us consider some graphical interpretations.

3.1. At first sight, those episodes cannot be captured by simple neoclassical models

A Neoclassical growth model

- We introduce the following setup:

$$E_0 \sum_{t=0}^{\infty} \beta^t U(C_t, 1 - L_t)$$

$$C_t + I_t = F(K_t, L_t)$$

$$K_{t+1} = I_t + (1 - \delta)K_t$$

3.1. At first sight, those episodes cannot be captured by simple neoclassical models

A Graphical Interpretation



$$C + I = F(K, L) \quad (GM)$$

$$U_1(C, 1 - L)F_2(K, L) = U_2(C, 1 - L) \quad (LM)$$

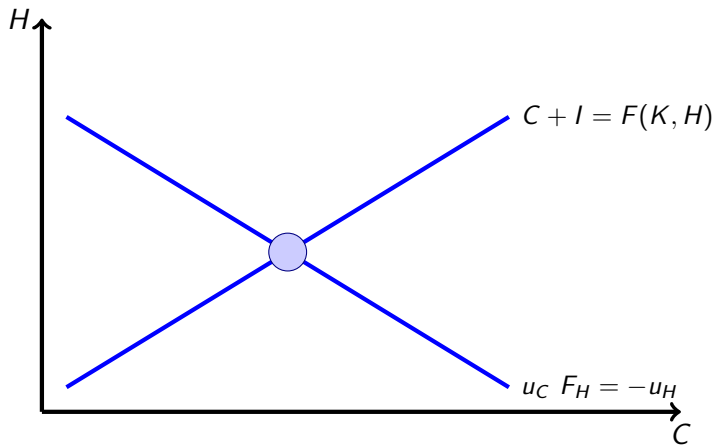
that defines the temporary equilibrium

- ▶ plus a Euler equation

$$U_1(C, 1 - L) = \beta EU_1(C', 1 - L')(1 + F_1(K', L') - \delta)$$

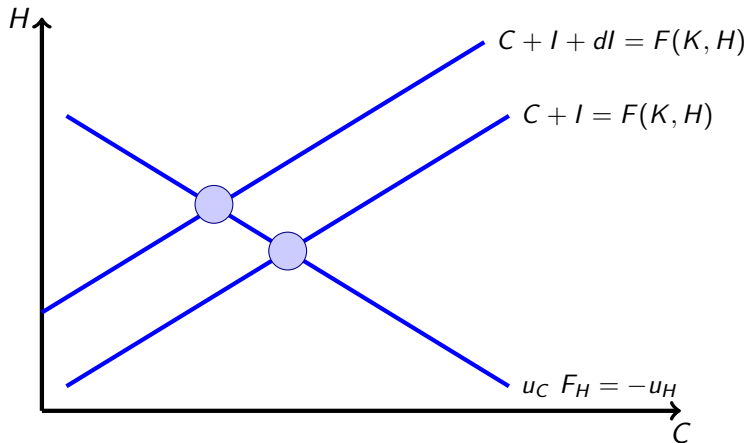
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A Graphical Interpretation



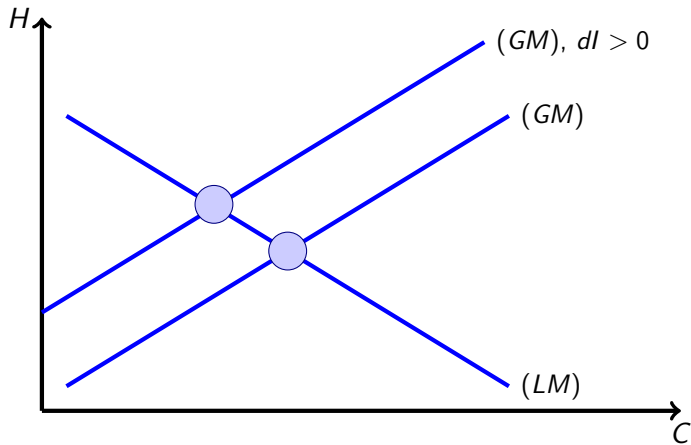
3.1. At first sight, those episodes cannot be captured by simple neoclassical models

A Graphical Interpretation



3.1. At first sight, those episodes cannot be captured by simple neoclassical models

A Graphical Interpretation



3.2. A Model with Increasing Returns

- In simple analytic version of Benhabib & Farmer:

$$U(C, L) = \log(C) + B(1 - L)$$

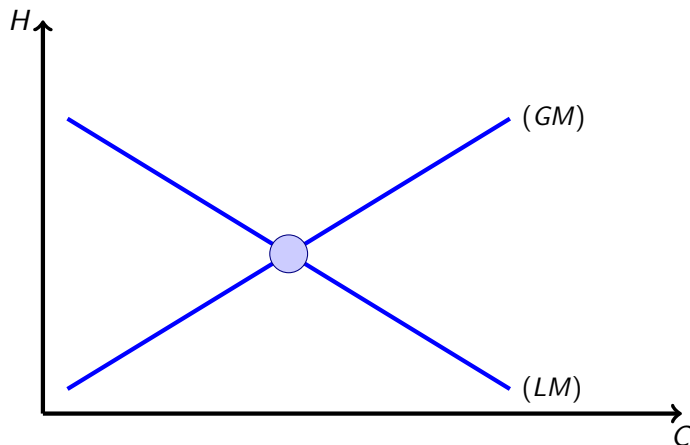
$$Y_i = K_i^\alpha L_i^{1-\alpha} \left(K^{\alpha'} L^\beta \right)$$

- The temporary equilibrium of this economy can be shown to be the triplet (C, L, I) that satisfies a RC and a LM equation

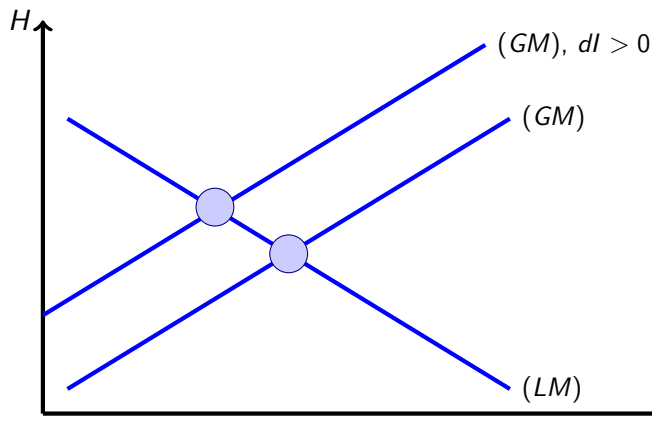
$$dC + dI = (1 - \alpha + \beta) \frac{Y}{L} dL \quad (GM)$$

$$dC = (\beta - \alpha) \frac{C}{L} dL \quad (LM)$$

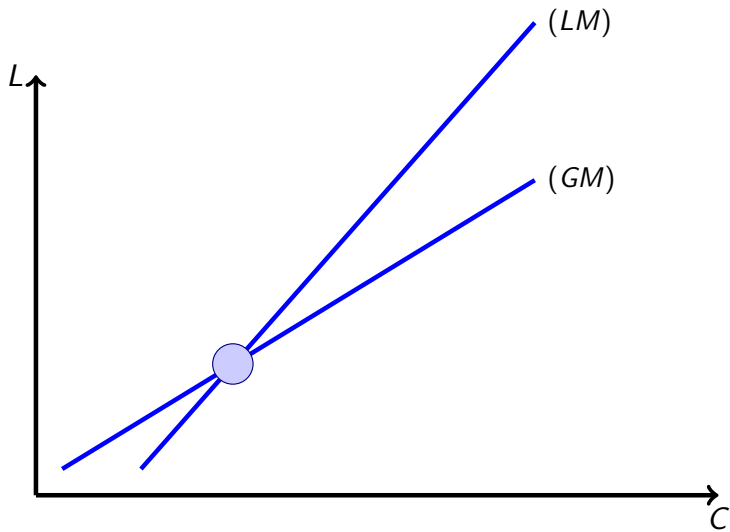
A Model with (not enough) Increasing Returns ($\beta < \alpha$)



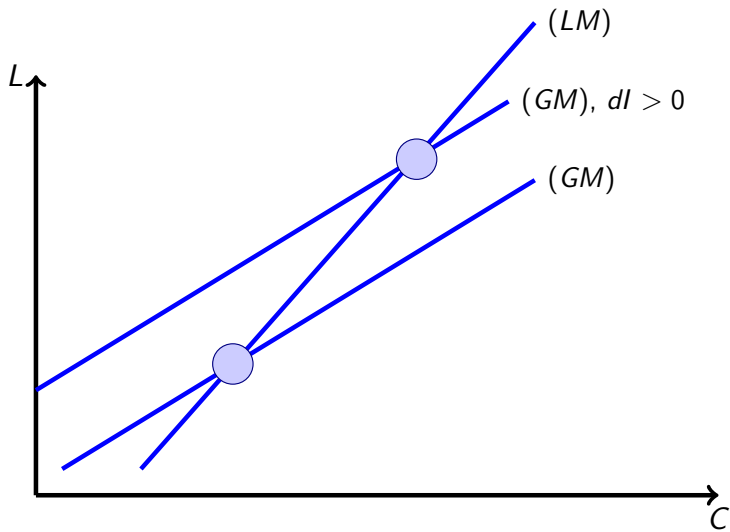
A Model with (not enough) Increasing Returns ($\beta < \alpha$)



A Model with (enough) Increasing Returns ($\beta > \alpha$)



A Model with (enough) Increasing Returns ($\beta > \alpha$)



A Sticky-Price Model (IS-LM)

Money cannot be carried over from one period to another, is distributed in every period to household in a lump-sum way. There is a transaction constraint such that only money and wage income can be used to buy consumption goods, and we assume that this constraint binds. In such a case, the household problem is given by

$$\begin{aligned} \max_{C,L,M} \quad & E_0 \sum_t \beta^t (\log(C_t) + B(1 - L_t)) \\ \text{s.t.} \quad & P_t C_t + P_t I_t + M_t^d \leq P_t w_t L_t + P_t z_t K_t + M_t \\ & K_{t+1} \leq (1 - \delta) K_t + I_t \\ & P_t C_t \leq M_t^d + P_t w_t L_t \end{aligned}$$

A Sticky-Price Model (IS-LM)

- If we assume competitive firm behavior, the flex-price temporary equilibrium allocations is given by:

$$C_t + I_t = K_t^\alpha L_t^{1-\alpha} \quad (GM)$$

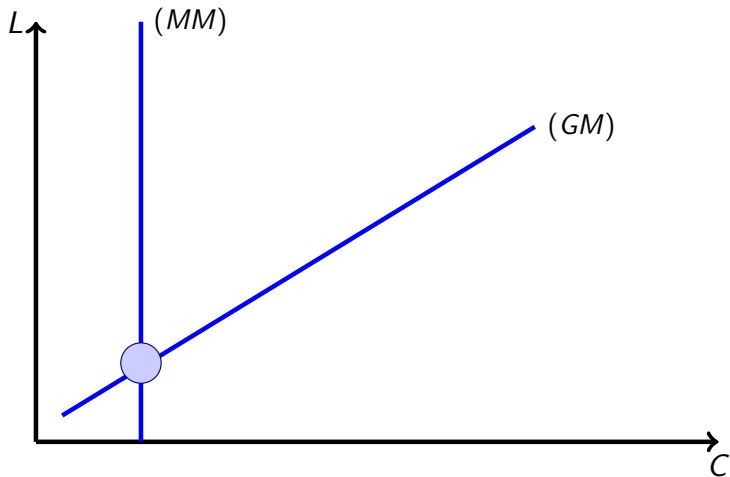
$$\frac{(1-\alpha)K_t^\alpha L_t^{-\alpha}}{C_t} = B \quad (LM)$$

$$\alpha C_t - (1-\alpha)I_t = \frac{M}{P_t} \quad (MM)$$

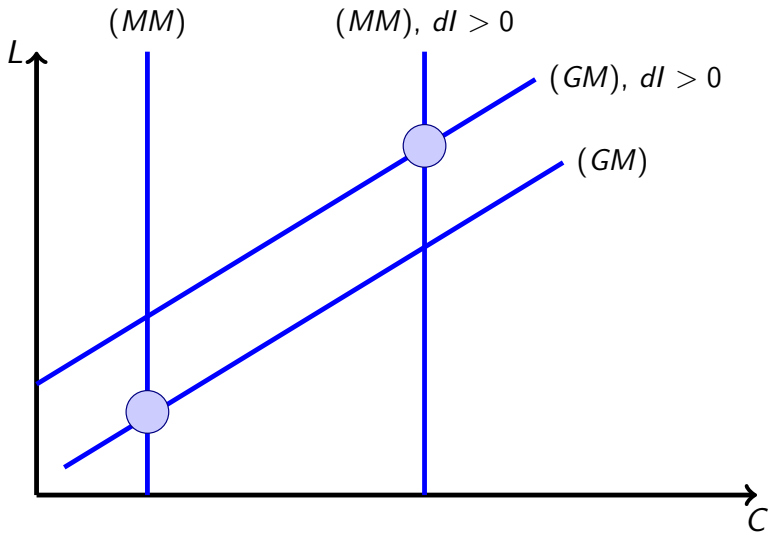
A Sticky-Price Model (IS-LM)

- ▶ We now assume that prices are preset (at their walrasian level),
- ▶ We also assume that labor market allocations are always determined by labor demand.
- ▶ Under those conditions, a fix-price equilibrium is a (C_t, L_t, I_t) that satisfies equations (GM) and (MM) , the labor market being generically not in equilibrium.
- ▶ (The temporary equilibrium is an IS-LM model)

A Sticky-Price Model (IS-LM)



A Sticky-Price Model (IS-LM)



Summary

Is that a case for non-walrasian features?

- ▶ No, as we show now.

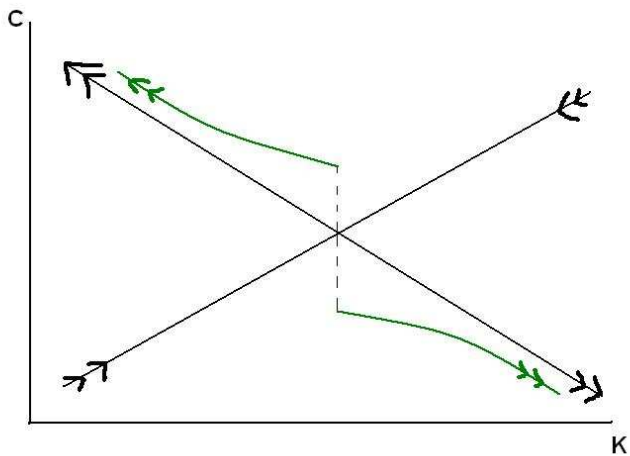
Outline of the Talk

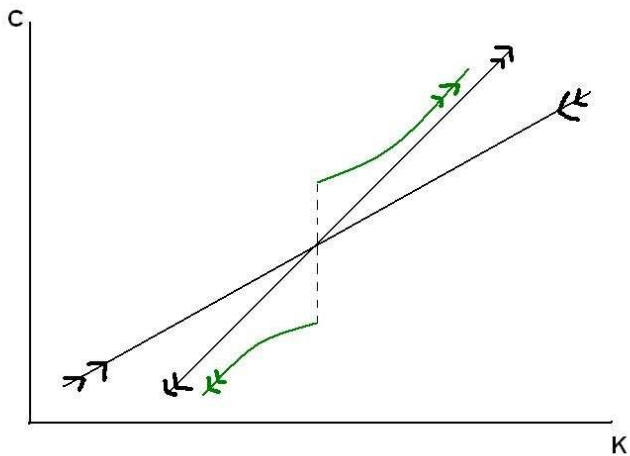
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4. Theory in simple models

4.1. Simple?

- ▶ Simple neo-classical setting: capital stock is the only state variable, and that technology and preferences are steady growth compatible.
- ▶ Our results thus have simple two dimensional phase diagram representation.





Comments

- ▶ Both stable arms are positively sloped $\rightsquigarrow$ same similar properties of the transitional dynamics (but not in growth rates)
- ▶ Very different behavior following expectations shocks.

4.2. Overview of the result we obtain (Beaudry and Portier jet-2006)

- ▶ *(i)* Expectation driven fluctuations are possible in simple neo-classical settings
- ▶ *(ii)* Not in the most commonly used macro models.
- ▶ *(iii)* Some internal cost complementarities are needed
- ▶ Not an exotic feature: implicitly implied by models which adopt inter-sectorial adjustment costs.

4.3 In a nutshell:

- We introduce the following setup:

$$E_0 \sum_{t=0}^{\infty} \beta^t U(C_t, 1 - L_t) \quad (3)$$

$$C_t = G(K_t, L_t, I_t; \psi_t) \quad (4)$$

$$K_{t+1} = I_t + (1 - \delta)K_t \quad (5)$$

4.3 In a nutshell:

$$C_t = G(K_t, L_t, I_t; \psi_t) \quad (4)$$

Remark : Equation (4) encompasses many of the macro models we use.

- ▶ One sector growth model $G(\cdot) = F(K_t, L_t) - I_t$, with $F(\cdot)$ concave
- ▶ Tobin's Q model $G(\cdot) = F(K_t, L_t) - I_t - \phi\left(\frac{I_t}{K_t}\right) K_t$, with $\phi(\cdot)$ a convex cost of adjustment function.

4.3 In a nutshell:

- ▶ Standard two-sector model: $G(\cdot)$ is the envelop function defined by

$$\begin{aligned} G(K, L, I) = \max_{K^c, K^I, L^c, L^I} \quad & F^c(K^c, L^c) \\ \text{s.t.} \quad & \\ & F^I(K^I, L^I) \geq I_t \\ & K^c + K^I \leq K_t \\ & L^c + L^I \leq L_t \end{aligned}$$

where $F^c(K^c, L^c)$ is the production function for consumption goods and $F^I(K^I, L^I)$ for investment goods.

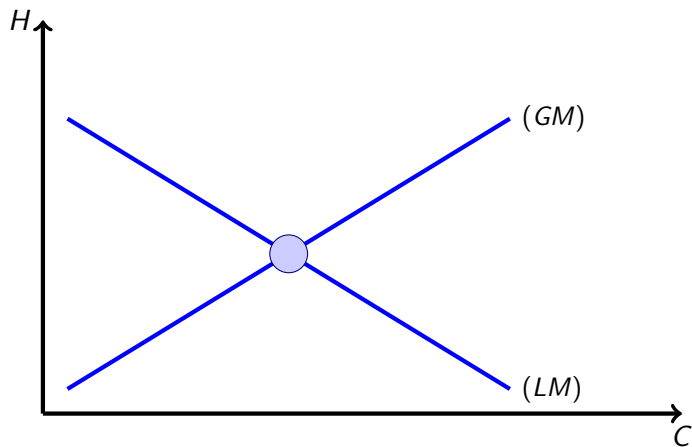
4.3 In a nutshell:

- ▶ We show that $G_{LI} > 0$ is a necessary condition for Expectation Driven Business Cycles (“Pigou Cycles”)
- ▶ Interpretation: cost complementarity.
- ▶ Labor is marginally more productive in the consumption good sector when the economy is investing a lot.
- ▶ Temporary equilibrium:

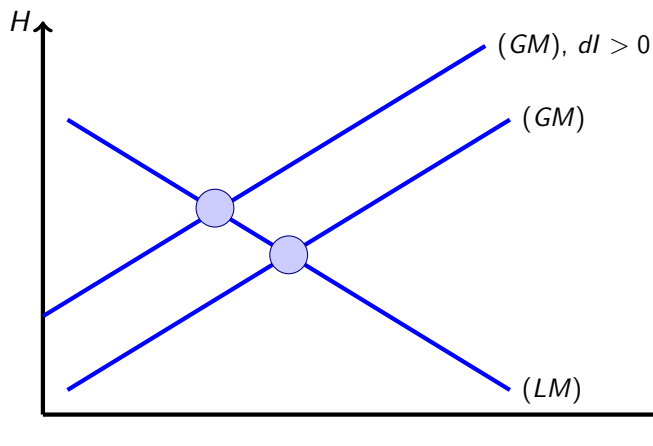
$$C = G(K, L, I) \quad (GM)$$

$$U_C(C, 1 - L)G_L(K, L, I) = -U_L(C, 1 - L) \quad (LM)$$

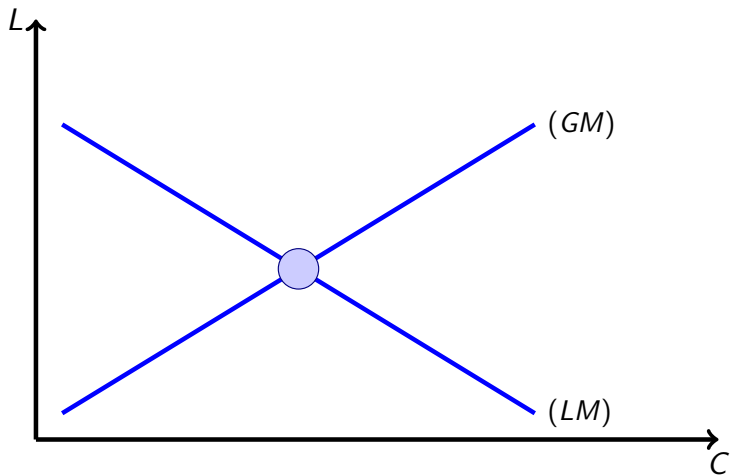
The Standard One-Sector Mode



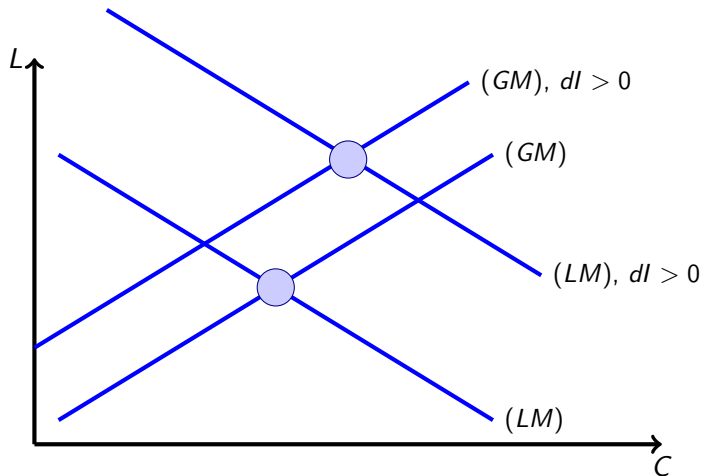
The Standard One-Sector Mode



A Model with (large enough) Positive G_{LI}



A Model with (large enough) Positive G_{LI}



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9. Fiscal Policy

5. Application 1: A theory of recessions

- ▶ A good news creates a boom.
- ▶ What if the news happen to be wrong? deep recession.
- ▶ Can the BC be explained by news and revisions about future technological progress?
- ▶ Beaudry & Portier jme-2004: yes, to some extent.

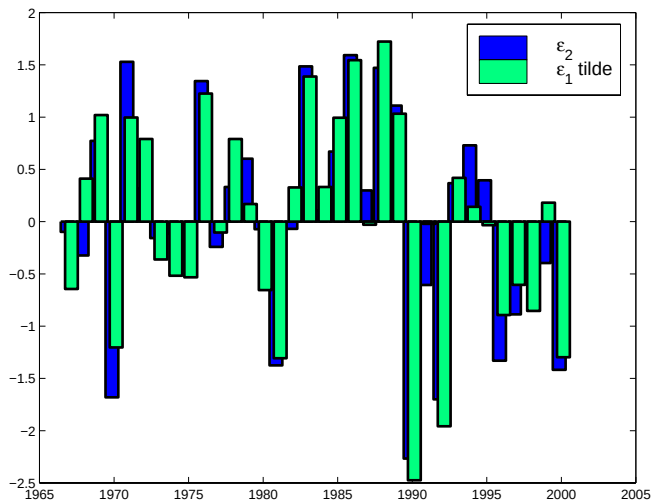
Other models

- ▶ Following Beaudry & Portier jme-2004, some scholars have proposed models that generate “Pigou cycles”.
- ▶ Deen Haan & Kaltenbrunner, Jaimovitch & Rebelo, Christiano, Motto & Rostagno, Mertens, ...
- ▶ Beaudry & Portier 2008 consider heterogenous agents, Beaudry, Collard & Portier 2008 look at news about future rents (gold rushes)

5.1. Japan Lost Decade (Beaudry & Portier 2004)

- ▶ How to explain the “bubble burst” of the early 90's, then the slow growth of the 90's
- ▶ A downward revision of TFP growth (that is left to be explained)
- ▶ This is what the data suggest.

Estimated ϵ_2 and $\tilde{\epsilon}_1$



5.1. Japan Lost Decade

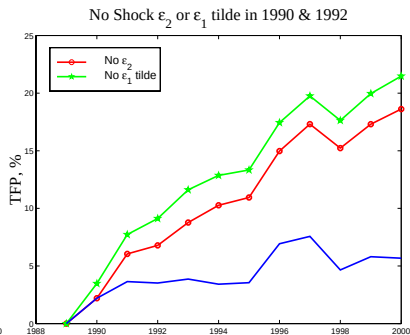
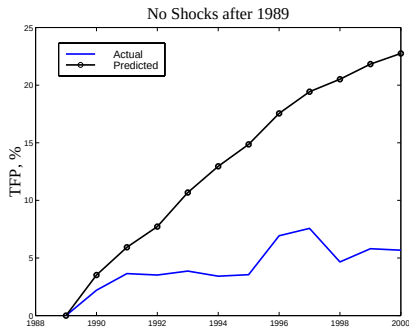
- ▶ What do we get from this figure?
- ▶ two stock market shocks at the beginning of the 1990s, that where possibly the consequence of bad news about future TFP , explain most of TFP changes in the 1990s and about half of the stock market variations.
- ▶ Gives some rationale to Hayashi and Prescott (2002) intuition:

5.1. Japan Lost Decade

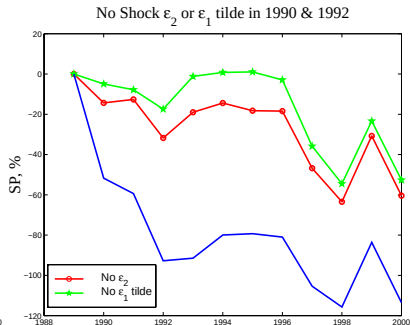
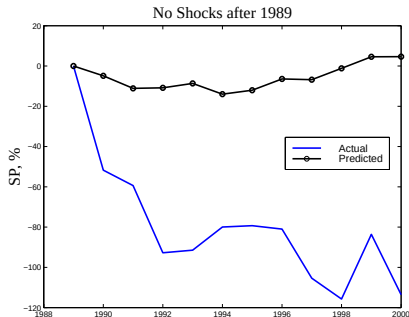
“ In examining the virtual stagnation that Japan began experiencing in the early 1990s, we find that the problem is not a breakdown of the financial system [...]. The problem is low productivity growth. [...] We said very little about the “bubble” period of the late 1980s and early 1990s, a boom period when property prices soared, investment as a fraction of GDP was unusually high, and output grew faster than in any other years in the 1980s and 1990s. We think the unusual pickup in economic activities, particularly investment, was due to an anticipation of higher productivity growth that never materialized. To account for the bubble period along these lines, we need to have a model where productivity is stochastic and where agents receive an indicator of future productivity.”
(italics added by myself)

(page 227-228), Hayashi & Prescott, *Japan's Lost Decade*, RED, 2002

Historical Decomposition of the 1990s



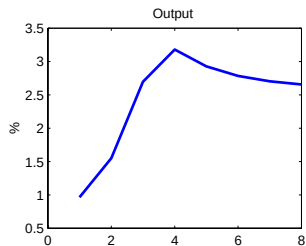
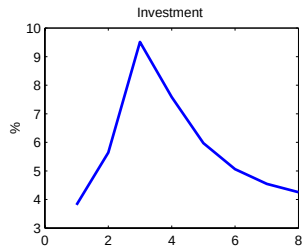
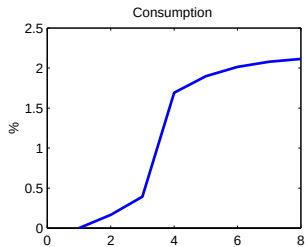
Historical Decomposition of the 1990s



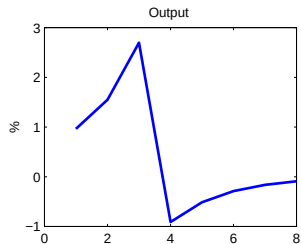
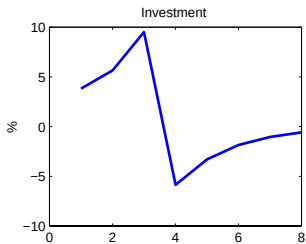
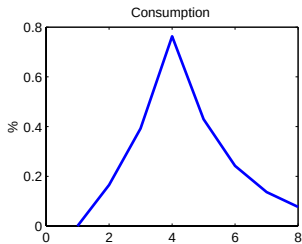
5.2. A Model of U.S. booms and recessions (Beaudry & Portier jme-2004)

- ▶ A model with two-sectors
- ▶ A model with noisy signals about future TFP growth
- ▶ Agents filter information, take decisions rationally, but are “wrong” from time to time.
- ▶ This creates fluctuations quite similar to actual ones.

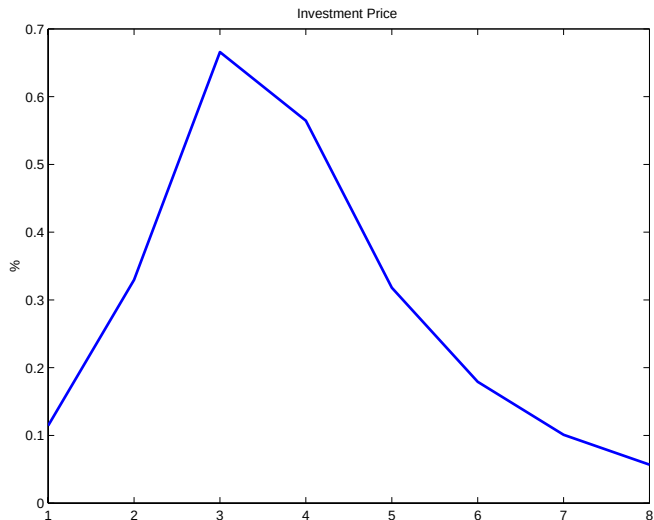
Realized News



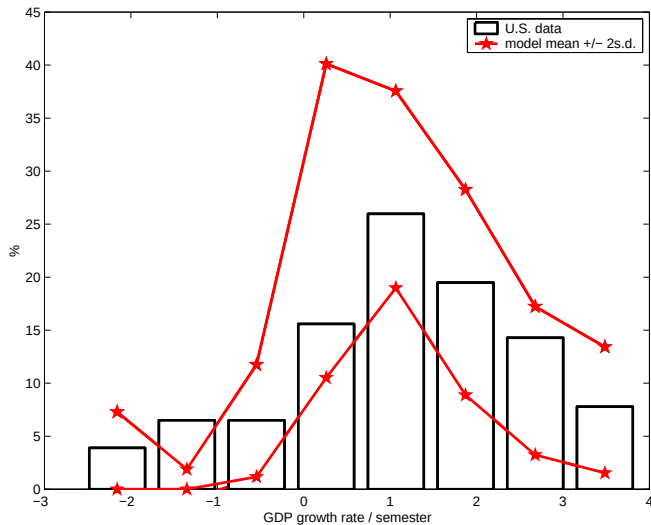
Not Realized News



The Behavior of Investment Price in a Pigou Cycle



Data and Model Expansions and Recessions



Outline of the Talk

1. Motivations
2. Facts
3. Why are those facts challenging for theory?
4. Theory in simple models
5. Application 1: A theory of recessions
6. Application 2: A theory of international comovements
7. Adding heterogeneity
8. Application 3: Non inflationary business cycles
9. Fiscal Policy

6. Application 2: A theory of international comovements

6.1. International Business Cycle Comovements

- ▶ At business cycle frequencies, Y , C , I , H are:
 - positively correlated within countries,
 - pairwise positively correlated among countries.
- ▶ Local technology shocks imply reallocation of mobile inputs
 $\rightsquigarrow$ negative comovements unless almost perfectly correlated shocks.
- ▶ As technology shocks are not 'global enough', 'Demand' shocks and frictions are needed for synchronization business cycles.

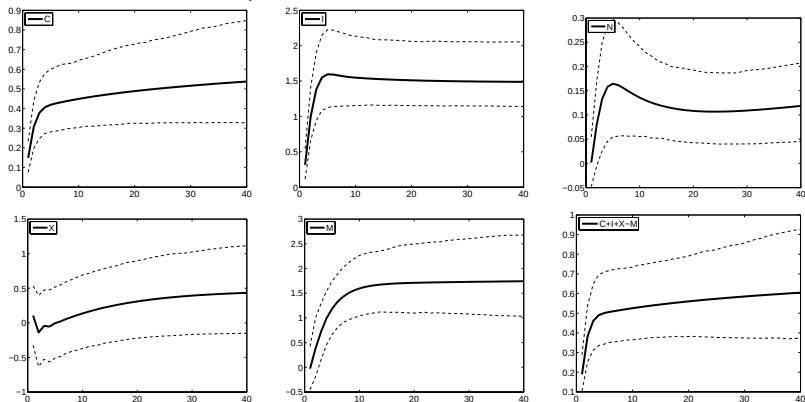
6.2. Our Contribution (Beaudry, Dupaigne & Portier, 2011)

- ▶ We show theoretically that shocks to expectations (news) are indeed creating international comovements. (*even if the realization is local, information is global*)
- ▶ We identify in the data a particular news (TFP news), and show that it is internationally transmitted (US/Canada, Germany/Austria).
- ▶ We propose a model that is able to replicate the qualitative features of the data.

6.3. The international propagation of news shocks

A news shock triggers an expansion in the US...

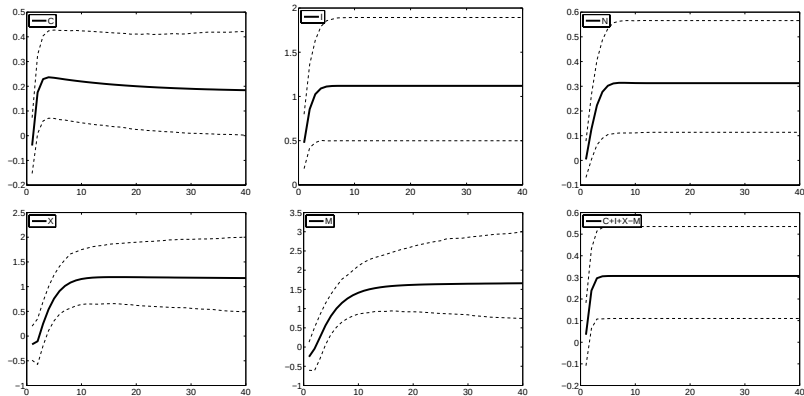
Response to a news shock, USA



6.3. The international propagation of news shocks

...as well as in Canada.

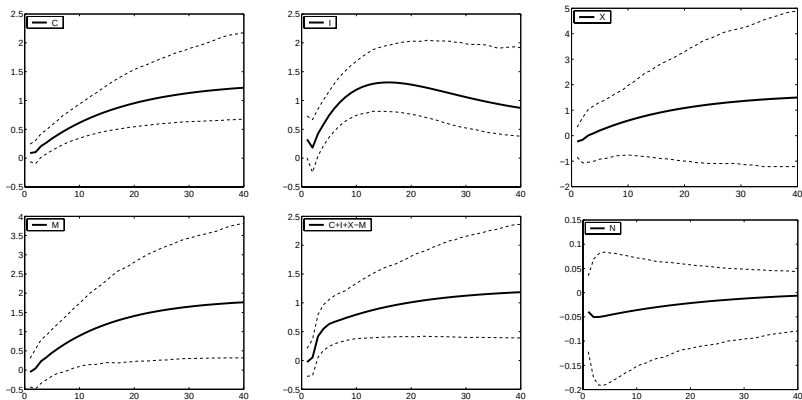
Response of Canadian aggregates to a news on US TFP



6.3. The international propagation of news shocks

Again a news shock triggers an expansion in Germany...

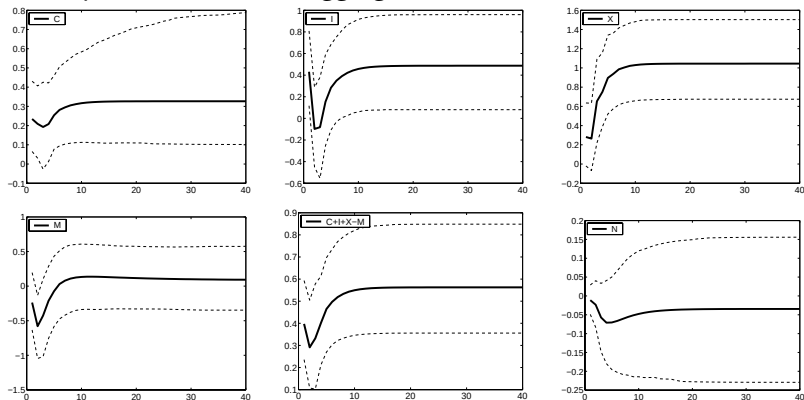
Response to a news shock, Germany



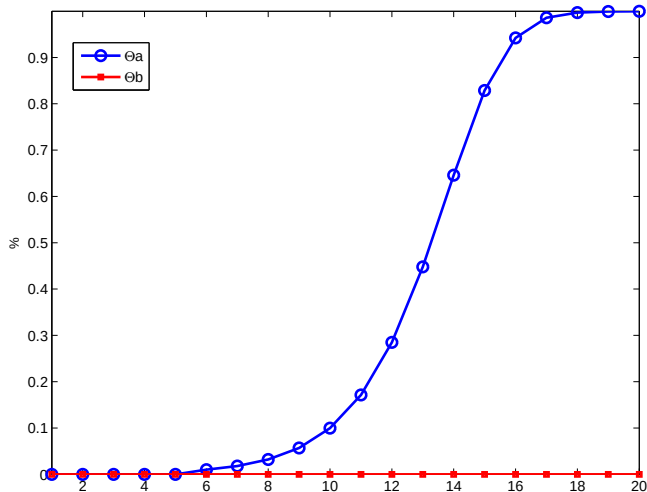
6.3. The international propagation of news shocks

... as well as in Austria.

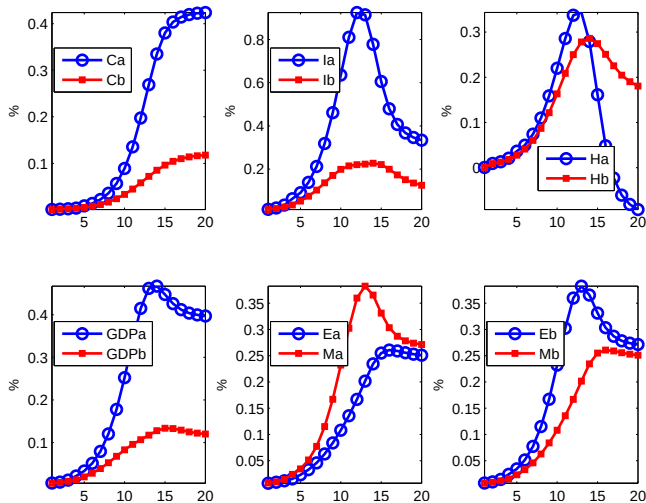
Response of Austrian aggregates to a German News Shock



6.4. The effects of country-A technological news in the model



6.4. The effects of country-A technological news in the model



6.4. The effects of country-A technological news in the model

- ▶ News shocks are powerful synchronizing forces, and can drive national and international business cycles in not-so-strange settings.
- ▶ Observed news shocks create comovements *across* and *within* countries.
- ▶ Demand shocks are needed in open macro: News are candidate in flex-price-complete markets environments.

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7.1 Framework

Setup

- ▶ Two-agents/two-sector economy

7.1. Framework

Preferences and Technology

- ▶ $U^i(C_i, 1 - L_i) + V^i(K_i; \Omega)$
- ▶ Concave CRS technologies in both sectors.
- ▶ $C = F^C(L_1^C, L_2^C)$ and $K = F^I(L_1^I, L_2^I)$.
- ▶ We will contrast:
 - ▶ integrated labor markets: $C = F^C(L_1^C + L_2^C)$ and $K = F^I(L_1^I + L_2^I)$.
 - ▶ (very) segmented labor markets: $C = F^C(L_1)$ and $K = F^I(L_2)$.

7.2. Perception Driven Fluctuations (News Shocks, etc)

- ▶ We study the competitive equilibrium of that economy.
- ▶ Under which conditions does a increase in the perceived marginal value of capital ($d\Omega_1 > 0$) does create a boom?
Does a decrease create a bust?

7.2. Perception Driven Fluctuations (News Shocks, etc)

Definitions

Definition 1 : **aggregate positive co-movement** : *aggregate consumption, aggregate investment and aggregate employment all strictly move in the same direction.*

Definition 2 : **individual positive co-movement**: *consumption, investment and employment levels for all individuals move in the same direction.*

7.2. Perception Driven Fluctuations (News Shocks, etc)

Walrasian equilibrium puts little restrictions on allocations

Proposition 1 : *Following a change in perceptions,*

- ▶ *Aggregate positive co-movements are possible,*
- ▶ *Individual co-movements are not possible.*

7.2. Perception Driven Fluctuations (News Shocks, etc)

The representative agent case

Corollary 1 : *With a representative agent, positive aggregate co-movement **are not possible**.*

7.2. Perception Driven Fluctuations (News Shocks, etc)

The importance of labor market segmentation

- ▶ What does matter for aggregate positive co-movements?
- ▶ Preference heterogeneity or labor market segmentation?

7.2. Perception Driven Fluctuations (News Shocks, etc)

The importance of labor market segmentation

Proposition 2 :

- ▶ *If labour markets are fully integrated, aggregate positive co-movement **are not possible**.*
- ▶ *If preferences are identical and labour markets not fully integrated, aggregate positive co-movement **are possible**.*

7.3. The Story

- ▶ Spain
- ▶ Two types of households
- ▶ Carpenters and Farmers
- ▶ Houses (capital good) and tomatoes (consumption good)
- ▶ In the short run, attachment to a sector

7.3. The Story

- ▶ The carpenter needs to eat, the farmer needs a shelter
- ▶ Static Gains from Trade between the two.
- ▶ Assume that the perceived value of houses decreases (bad news, pessimism)
- ▶ The relative price of houses in terms of tomatoes p will go down
- ▶ The carpenter will work less for two reasons
 1. he does not want as many houses as before
 2. he cannot trade as many houses as before
- ▶ L_I and I will go down.
- ▶ The farmer does not want to buy as many houses as before, and does not need to produce as many tomatoes for the trade: L_C and C .

7.3. The Story

- ▶ C , I , L in both sectors, $Y = C + pI$ are all going down
- ▶ If reallocation of workers take some time, the recession is likely to be protracted.
- ▶ Changes in perceptions about the future are affecting the depth of Gains from Trade today
- ▶ Fluctuations are here related with variations in the amount of Gains from Trade.

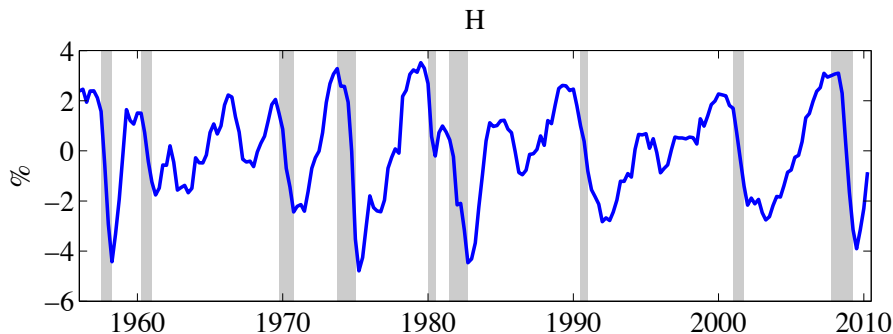
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8.1. Some Intriguing Facts over the last 3 cycles

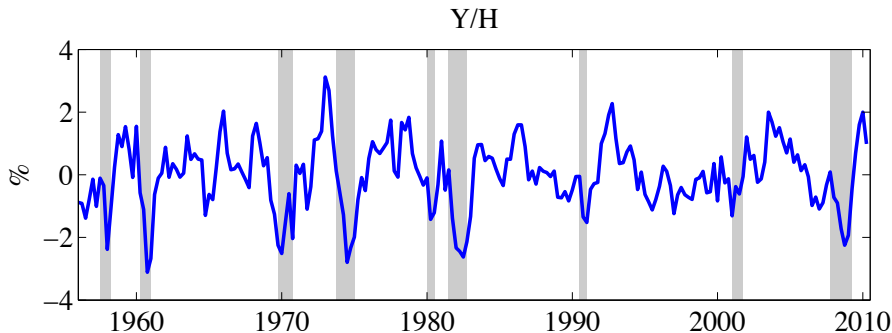
- ▶ Over the last 30, the US economy (can be extended to the Eurozone) has experienced an intriguing type of business cycle.
- ▶ Let's look at
 - ▶ HP filtered hours H
 - ▶ HP filtered average labor productivity Y/H
 - ▶ Raw core inflation

8.1. Some Intriguing Facts over the last 3 cycles



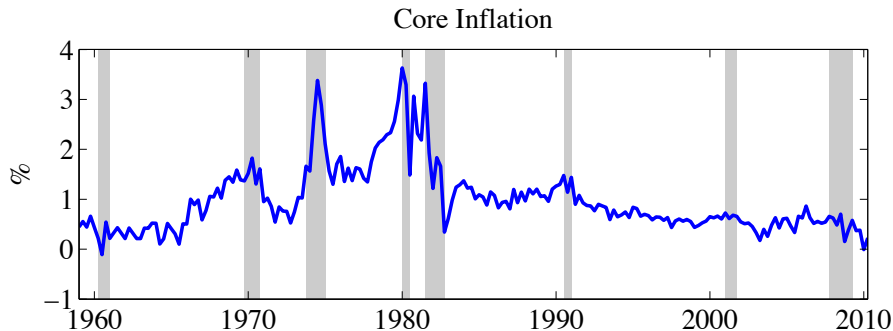
- ▶ The last (say) 3 business cycles do not look that different from the ones before.

8.1. Some Intriguing Facts over the last 3 cycles



- ▶ Productivity has been (at best) a-cyclical.
- ▶ This is pointing towards “demand” shocks as generating recent BCs.

8.1. Some Intriguing Facts over the last 3 cycles



- ▶ Core-inflation has been very stable.
- ▶ Almost no variance nor co-variance with output.

8.2. The Trouble with RBC Models

- ▶ This pattern is hard to explain in the RBC tradition :
- ▶ No procyclical TFP shocks.
- ▶ Demand shocks typically cause negative consumption - investment co-movements.

8.3. The Trouble with NK Models

- ▶ If output moves because the output gap moves (demand shocks), then we should expect inflation response.
- ▶ The core NK model has more than “troubles”
- ▶ Take the Jordi Galí’s textbook New Phillips curve

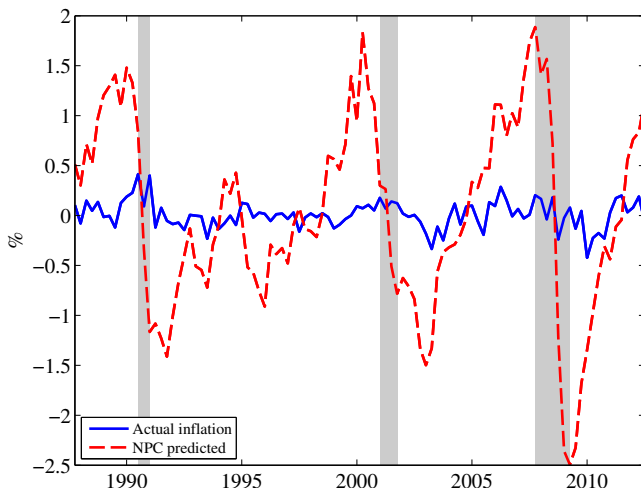
$$\pi_t = \beta E_t \pi_{t+1} + \kappa \tilde{y}_t + u_t$$

- ▶ Assume that the output gap is AR(1) with persistence ρ .

$$\pi_t = \frac{\kappa}{1 - \beta\rho} \tilde{y}_t + u_t$$

- ▶ Take Jordi’s textbook calibration (including a mean duration of prices of 3 quarters).
- ▶ Assume that the output gap is measured by the HP cycle of output (conservative assumption).
- ▶ Feed it into this last equation and deduct the implied inflation, killing cost-push shocks.

8.3. The Trouble with NK Models



- Then NPC implies that s.d. of inflation is **3** times larger than in the data

8.4. Introducing “Gains from Trade”

- ▶ Introduce our limited mobility / two sector framework within an otherwise standard New Keynesian model.
- ▶ Consider believes that future returns to capital will be high.
- ▶ Consumption producing agents want to by capital as it return is expected to be high.
- ▶ They will trade more with investment producing agents.
- ▶ Investment producing agents will also want more capital, but also more consumption because of an income effect.
- ▶ This would look like a demand shock.
- ▶ But it does not put any pressures on prices, as the natural or non-inflation rate of output has also changed.
- ▶ Same thing with downward revisions: recession but no deflation.

8.4. Introducing “Gains from Trade”

- ▶ We think it is a useful model to think of the last episodes of expansions and recessions.
- ▶ Productivity is not procyclical
- ▶ Inflation barely moves
- ▶ Suggest that booms and busts are mainly driven by expectations on investment return.

8.5. The basic model

- ▶ Let's start with the textbook New Keynesian model.
- ▶ $c_t = \left(\int_0^1 c_{jt}^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}}$
- ▶ $\sum \beta^t (\ln(c_{ct}) + \Phi(1 - \ell_{Ct}))$.
- ▶ Monopoly $j : C_{jt} = A_t L_{jt}$
- ▶ Calvo price setting
- ▶ Monetary authorities follow a Taylor rule.

8.5. The basic model

The basic model

- ▶ This basic model collapses to a Phillips curve and a dynamic IS equation + Taylor rule

$$\left\{ \begin{array}{lcl} \tilde{y}_t & = & -(\hat{i}_t - E_t \hat{\pi}_{t+1} - \hat{\rho}_t^n) + E_t \tilde{y}_{t+1} \\ \hat{\pi}_t & = & \beta E_t \hat{\pi}_{t+1} + \lambda (\hat{y}_t - \hat{y}_t^n) \\ + & & \text{Taylor rule} \end{array} \right.$$

- ▶ $\hat{y}_t^n = \hat{A}_t$

8.5. A GFT approach

The basic model

$$\begin{cases} \tilde{y}_t &= -(\hat{v}_t - E_t \hat{\pi}_{t+1} - \hat{\rho}_t^n) + E_t \tilde{y}_{t+1} \\ \hat{\pi}_t &= \beta E_t \hat{\pi}_{t+1} + \lambda (\hat{y}_t - \hat{y}_t^n) \\ + &\text{Taylor rule} \end{cases}$$

- ▶ What happens in this environment if agents expect $\hat{A}_{t+1}$ to be high?
- ▶ It is a demand shock to current consumption.
- ▶ Absent of an active Taylor Rule, inflation.
- ▶ The increased expectation of A_{t+1} does not directly enter into the Phillips curve.
- ▶ Therefore, if it creates an output boom, it will create inflation (no change in the natural output)

8.5. A GFT approach

The basic model

$$\left\{ \begin{array}{lcl} \tilde{y}_t & = & -(\hat{v}_t - E_t \hat{\pi}_{t+1} - \hat{\rho}_t^n) + E_t \tilde{y}_{t+1} \\ \hat{\pi}_t & = & \beta E_t \hat{\pi}_{t+1} + \lambda (\hat{y}_t - \hat{y}_t^n) \\ + & & \text{Taylor rule} \end{array} \right.$$

- ▶ What happens in this environment if agents observe high $\hat{A}_t$?
- ▶ It is a supply shock .
- ▶ It shows up directly in the the Phillips curve ($\hat{y}_t^n = \hat{A}_t$).
- ▶ Therefore, if it creates an output boom, it will not create inflation (changes in the natural output)

8.5. A GFT approach

The basic model

- ▶ Textbook prescription: to keep stable inflation, monetary authorities need to strongly counteract demand shock but need to accommodate supply shocks.
- ▶ If the economy goes into recession due to a fall in demand – as opposed to a reduction in supply capacity – this should put substantial downward pressure on prices.
- ▶ Let's contrast those results with the one of a gains-from-trade model.

8.5. A GFT approach

The gains-from-trade model

- ▶ Add a mass 1 of agents (type 2)
- ▶ Type 2 individuals produce some capital good (full depreciation): $K_{t+1} = L_{2t}$.
- ▶ Preferences: $\ln \left(C_{2t} + \frac{\psi_2}{2}(1 - L_{2t})^2 \right)$.
- ▶ Capital market is flex-price and competitive.
- ▶ Production of consumption good: $C_{jt} = \Theta_t K_{jt} + L_{1jt}$
- ▶ If then mass of type 2 individuals is 0, then there is no capital supply and the model is the previous one.

8.5. A GFT approach

The gains-from-trade model

- ▶ The log linear approximation of the model is

$$\left\{ \begin{array}{lcl} \tilde{y}_t & = & -\zeta (\hat{i}_t - E_t \hat{\pi}_{t+1} - \hat{\rho}_t^n) + E_t \tilde{y}_{t+1} \\ \hat{\pi}_t & = & \beta E_t \hat{\pi}_{t+1} + \lambda \zeta^{-1} (\hat{y}_t - \hat{y}_t^n) \\ + & & \text{Taylor rule} \end{array} \right.$$

- ▶ Natural or non-inflationary level of output

$$\hat{y}_t^n = \phi_2 A_t + \phi_1 E_t [\hat{\Omega}_{t+1}] \text{ where } \hat{\Omega}_{t+1} = \hat{\Theta}_{t+1} - \hat{A}_{t+1}$$

8.5. A GFT approach

A new new Phillips curve

$$\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + \lambda \zeta^{-1} \left(\hat{y}_t - \phi_2 A_t - \phi_1 E_t \left[\hat{\Omega}_{t+1} \right] \right)$$

- ▶ Consider believes that Ω_{t+1} will be high.
- ▶ Agents of type 1 to want to by capital as it return is expected to be high.
- ▶ They will trade more with agents of type 2.
- ▶ Type 2 agents will also want more capital, but also more consumption because of an income effect.
- ▶ This would look like a demand shock.
- ▶ But it does not put any pressures on prices, as the natural or non-inflation rate of output has also changed.
- ▶ Same thing with downward revisions: recession but no deflation.

8.6. Evidence of Labour Market Segmentation and Imperfect Insurance

Main idea

- ▶ Our theory relies on two main features of the world
 - ▶ (1) Sluggish sectoral reallocation of labor
 - ▶ If not, perceived changes in the value of capital would cause reallocations, not changes in the depth of gains to trade.
 - ▶ (2) Imperfect risk sharing
 - ▶ If not, consumption is immune to changes in expectations
- ▶ Is it possible to test for this dependence using PSID
- ▶ Idea: under full risk sharing and labor mobility, the beginning of period t sector of employment of an individual should not influence, *ceteris paribus*, the growth of her wage income or consumption between, say, t and $t + n$.

8.6. Evidence of Labour Market Segmentation and Imperfect Insurance

Data

- ▶ Data from the PSID over the period 1968-2007 (yearly until 1997, bi-annual since).
- ▶ We observe for the family head and for every family-year
 - ▶ sector of activity at the beginning of the period
 - ▶ labour income
 - ▶ food consumption
 - ▶ age, education
- ▶ We also observe employment and wage bill at the sectoral level (using NIPA)

8.6. Evidence of Labour Market Segmentation and Imperfect Insurance

Specification

- ▶ We estimate:

$$\Delta X_{it} = \text{time dummies} + \alpha_1 \Delta Emp_{j(i)t} \\ + \text{age dummies} + \text{sector dummies} + \epsilon_{it}$$

- ▶ X is first labor income, then consumption.
- ▶ $j(i)$ is the sector in which i was employed at the beginning of the period.
- ▶ Δ is taken over 2 years, or 1 year after 1997
- ▶ Other controls: highest level of educational attainment, and interactions between age-time, education-age, education-time.
- ▶ If labour markets were completely integrated, and given we are controlling for common time effects then individual level outcomes should not be systematically related to aggregate outcomes for any particular sectors.

8.6. Evidence of Labour Market Segmentation and Imperfect Insurance

Table 1: Effect of Sectoral Growth on Individual Income

	2-year	2-year	2-year	2-year	2-year	1-year	1-year	1-year
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ Emp	.542 (.209)		.468 (.244)					
Δ W-bill		.525 (.175)						
Δ Emp-10				.450 (.143)	.563 (.131)			
Δ Emp						.535 (.170)	.579 (.193)	
Δ Emp-10								.471 (.059)
Obs.	49338	49338	45469	45430	23173	68863	63677	61224
R^2	.028	.028	.028	.027	.026	.017	.018	.018

8.6. Evidence of Labour Market Segmentation and Imperfect Insurance

Table 1: Effect of Sectoral Growth on Individual Income

- ▶ Compare two individuals that were initially attached to two different sectors,
- ▶ assume one of the two sector grew by an extra 1% over two years,
- ▶ the individual initially attached to that sector saw his labour income grow by an additional .5% compared to the other individual.

8.6. Evidence of Labour Market Segmentation and Imperfect Insurance

Table 2: Effect of Sectoral Growth on Household Consumption (Food)

	2-year (1)	2-year (2)	2-year (3)	2-year (4)	2-year (5)	1-year (6)	1-year (7)	1-year (8)
Δ Emp	.268 (.092)		.267 (.104)					
Δ W-bill		.236 (.078)						
Δ Emp-10				.143 (.052)	.112 (.053)			
Δ Emp						.200 (.118)	.274 (.129)	
Δ Emp-10								.208 (.077)
Obs.	67758	67758	63686	52270	26898	89008	83942	65503
R^2	.014	.014	.013	.016	.015	.005	.005	.006

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9. Fiscal Policy

Gains from trade as a foundation for multipliers

- ▶ Typical story for multipliers in keynesian-cross-like-models:
 - ▶ Initial increase in demand for one good,
 - ▶ This increase in demand is met by an increase in production of that good, and therefore by an increase in income of those producing that good
 - ▶ Those agents then in turn increase demand for some goods, etc...
- ▶ This is not (at all) the mechanism we find in sticky prices models of the business cycle.
- ▶ Let us show that multipliers can be understood as the consequence of changes in the scope for gains to trade between agents.

9. Fiscal Policy

Framework

- ▶ We again use the simple model ($U^i(C_i, 1 - L_i)$ and $V^i(K_i; \Omega_1, \bar{K})$, two sectors)
- ▶ We abstract for Ricardian issues, and look at balanced-budget policies of the type

$$G^C + G^K = T_1 + T_2$$

- ▶ Under what conditions (if any) can a temporary increase in government spending create a multiplier effect?
- ▶ It is helpful to begin by focusing on government expenditures that are directed only at one sector.

$$G^K = T_1 + T_2$$

9. Fiscal Policy

Two questions

- ▶ Given a policy (G^K, T_1) (T_2 is then obtained from the govt BC), two questions can be asked:

Question 1 : amplification effect: *Can an increase in government purchases of capital good cause private agents to buy more capital goods?*

Question 2 : spillover effect: *Can an increase in government purchases of capital good cause private agent to consume more consumption goods?*

9. Fiscal Policy

Two propositions

Proposition 5 :

- ▶ *If same preferences and perfectly substitutable labour,*
- ▶ *no increase in C (spillover) nor I (amplification) are possible.*

9. Fiscal Policy

Two propositions

Proposition 6 :

- ▶ *If the conditions of Proposition 3 are met,*
- ▶ *C and total I will increase.*
- ▶ *However, no amplification (private I decreases)*

9. Fiscal Policy

Two propositions

- ▶ An increase in government purchases of capital shifts out the aggregate demand for capital, increasing the aggregate quantity purchased and increasing the price of capital.
- ▶ Individual demand curves for capital tend to shift in because of the tax increases.
- ▶ No amplification effect as the equilibrium price of capital increase and agents have downward sloping demand curves.
- ▶ Agents in the capital sector are getting higher income which leads them to want to consume more. This is where the gains from trade arise.
- ▶ The producers of consumption goods are willing to trade with the producers of capital goods as the added income for them can be used to pay taxes.

9. Fiscal Policy

Two propositions

- ▶ Could it be the case that the income effect on the producers of capital is great enough to lead such producers to actually substantially increasing their purchase of capital and create amplification?
- ▶ It can happen, but not under the conditions of Proposition 3 (aggregate capital demand is downward sloping).

9. Fiscal Policy

An Example

- ▶ We keep the same two periods environment:
In $C^i + \phi(1 - I^i) + \ln K^i$, $C = L_1$, $K = L_2$, P the relative price of capital.
- ▶ Fiscal policy: spend α on K (amount spent $\frac{\alpha}{P}$), $1 - \alpha$ on C
- ▶ Fiscal policy: tax β on type 2 agent, $1 - \beta$ on type 1.
- ▶ In equilibrium:
 - ▶ $GDP = \frac{4}{\phi}$ without spending
 - ▶ $GDP = \frac{4}{\phi} + 1 + 2(\alpha - \beta)$ when spending 1.
- ▶ Multiplier = $1 + 2(\alpha - \beta)$

Conclusion

- ▶ Lot of things to do : asset pricing, endogenize the acquisition and filtering of information
- ▶ À suivre...